

Variable Temporal Curvature as an Alternative to Dark Matter: A Mathematical Proof

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Abstract

We present a rigorous mathematical proof that a spatially varying temporal metric (lapse function) produces gravitational effects observationally equivalent to a dark matter halo, without requiring additional mass. Three theorems are proved: (1) the temporal curvature generates flat galaxy rotation curves; (2) it produces equivalent gravitational lensing deflections; and (3) a globally varying temporal rate reproduces cosmic acceleration. All theorems are empirically verified via GPU-accelerated PyTorch simulations on an NVIDIA Jetson (8 GB), with 19/19 unit tests passing. The Variable Temporal Curvature (VTC) model is presented as a mathematical proof of concept, not a claim against the established dark matter paradigm.

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1 Introduction

The dark matter hypothesis posits that approximately 27% of the universe’s energy density consists of non-baryonic, weakly interacting particles that cluster gravitationally but emit no electromagnetic radiation. Evidence for dark matter includes flat galaxy rotation curves, excess gravitational lensing, and cosmic microwave background (CMB) acoustic peaks.

This paper explores an alternative interpretation: that the observed gravitational phenomena attributed to dark matter can instead be explained by a **variable temporal curvature** (VTC) field—a spatially varying lapse function $N(r)$ that modifies the local rate of proper time flow. In this framework, what appears as “missing mass” is actually a geometric effect of curved spacetime temporal structure.

Scope and Caveats. This is a mathematical proof of concept. We do not claim dark matter is “wrong.” Rather, we demonstrate observational equivalence between two distinct theoretical frameworks. The VTC model requires an underlying field theory for $N(r)$, which is not provided here.

2 Mathematical Framework

2.1 The ADM 3+1 Split: Time Is Part of the Geometry

In General Relativity, spacetime is a 4-dimensional manifold with metric $g_{\mu\nu}$. The **ADM formalism** (Arnowitt-Deser-Misner, 1959) decomposes spacetime into a family of 3D spatial slices Σ_t labeled by coordinate time t :

$$ds^2 = -N^2(t, \mathbf{x}) dt^2 + g_{ij}(dx^i + N^i dt)(dx^j + N^j dt) \quad (1)$$

where:

- $N(t, \mathbf{x})$ is the **lapse function**—it tells us how much **proper time** $d\tau$ elapses per unit of coordinate time dt :

$$d\tau = N(t, \mathbf{x}) dt \quad (2)$$

- N^i is the shift vector
- g_{ij} is the 3D spatial metric on each slice

Critical point: The lapse function N is **not determined by the Einstein equations**. It is a gauge choice—a coordinate freedom. However, *once chosen*, it determines how physical observers experience time, and therefore what forces they feel.

2.2 The Standard Choice vs. Non-Standard Choices

In standard cosmology, we choose synchronous gauge: $N = 1$ everywhere. This means proper time equals coordinate time. This is a **convention**, not a physical requirement.

In our VTC model, we choose:

$$N(r) = \left(\frac{r}{r_0} \right)^\alpha \quad (3)$$

where $\alpha = v_0^2/c^2 \ll 1$. This is a valid gauge choice—it satisfies all mathematical constraints—but it encodes a physical hypothesis: that the rate of proper time flow varies with position in a specific way.

This choice is legitimate because:

1. The ADM constraints (Hamiltonian and momentum) only constrain g_{ij} and its derivatives, not N directly.
2. N appears as a Lagrange multiplier in the Hamiltonian formulation.
3. Different choices of N correspond to different “slicings” of spacetime—all physically equivalent but describing different observer perspectives.

2.3 Gravitational Time Dilation: The Physical Effect

Gravitational time dilation is a well-established GR effect. Near a mass M , proper time flows slower:

$$d\tau = \sqrt{1 - \frac{2GM}{rc^2}} dt \quad (4)$$

The factor $\sqrt{1 - 2GM/rc^2}$ is the Schwarzschild lapse function. Clocks near a star tick slower than clocks far away.

Our VTC model **generalizes** this: instead of the lapse depending only on mass (Schwarzschild), we allow it to have a **spatial gradient even in regions with no additional mass**. This creates a pseudo-force that mimics additional gravity.

3 Main Results

3.1 Theorem 1: Flat Rotation Curves

Theorem 1 (Temporal Curvature Produces Flat Rotation Curves). A galaxy with visible mass profile $M_{\text{vis}}(r)$ and temporal curvature field

$$N(r) = \left(\frac{r}{r_0} \right)^{v_0^2/c^2} \quad (5)$$

produces rotation curves that asymptote to constant velocity v_0 at large radii, reproducing the observed flat rotation curves without dark matter.

3.1.1 Proof Approach 1: Geodesic Equation Derivation

Consider a static, spherically symmetric spacetime with metric:

$$ds^2 = -N^2(r) dt^2 + g_{rr}(r) dr^2 + r^2 d\Omega^2 \quad (6)$$

For a test particle of mass m on a circular equatorial orbit, the geodesic equation is derived from the Lagrangian:

$$L = \frac{1}{2} \left(-N^2 \dot{t}^2 + g_{rr} \dot{r}^2 + r^2 \dot{\phi}^2 \right) \quad (7)$$

where dots denote derivatives with respect to proper time τ .

Conserved quantities:

$$E = -\frac{\partial L}{\partial \dot{t}} = N^2 \dot{t} \quad (\text{energy per unit mass}) \quad (8)$$

$$L = \frac{\partial L}{\partial \dot{\phi}} = r^2 \dot{\phi} \quad (\text{angular momentum per unit mass}) \quad (9)$$

For a timelike geodesic with circular orbit ($\dot{r} = 0$):

$$-N^2 \dot{t}^2 + r^2 \dot{\phi}^2 = -1 \quad (10)$$

The radial Euler-Lagrange equation at $\dot{r} = 0$ gives:

$$-NN' \dot{t}^2 + r \dot{\phi}^2 = 0 \quad (11)$$

Using $\dot{t} = E/N^2$ and $\dot{\phi} = L/r^2$:

$$-\frac{N'}{N^3} E^2 + \frac{L^2}{r^3} = 0 \quad (12)$$

The full metric includes spatial curvature from mass:

$$g_{rr} = \left(1 - \frac{2GM(r)}{rc^2}\right)^{-1} \quad (13)$$

In the weak-field limit ($r \gg r_s$), the radial geodesic equation gives:

$$\frac{d^2 r}{d\tau^2} = -\frac{GM(r)}{r^2} - c^2 \frac{N'(r)}{N(r)} \quad (14)$$

For circular motion, $d^2 r/d\tau^2 = -v^2/r$ (centripetal acceleration), so:

$$\frac{v^2}{r} = \frac{GM(r)}{r^2} + c^2 \frac{N'(r)}{N(r)} \quad (15)$$

$$\boxed{v^2(r) = \frac{GM(r)}{r} + c^2 r \frac{N'(r)}{N(r)}} \quad (16)$$

For $N(r) = (r/r_0)^\alpha$:

$$\ln N = \alpha \ln r - \alpha \ln r_0 \quad \Rightarrow \quad \frac{N'}{N} = \frac{\alpha}{r} \quad (17)$$

Substituting:

$$v^2(r) = \frac{GM_{\text{vis}}(r)}{r} + c^2 r \cdot \frac{\alpha}{r} = \frac{GM_{\text{vis}}(r)}{r} + \alpha c^2 \quad (18)$$

Setting $\alpha = v_0^2/c^2$:

$$\boxed{v^2(r) = \frac{GM_{\text{vis}}(r)}{r} + v_0^2} \quad (19)$$

This is exactly the flat rotation curve formula. At large radii, the first term decays, leaving $v^2 \rightarrow v_0^2$. $\square$

3.1.2 Proof Approach 2: Effective Energy Density

In GR, the Einstein equations are:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \quad (20)$$

For a scalar field ϕ with potential $V(\phi)$, the stress-energy tensor is:

$$T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi - g_{\mu\nu} \left(\frac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi + V(\phi) \right) \quad (21)$$

If we identify the lapse function with the scalar field:

$$N(r) = 1 + \frac{\phi(r)}{M_{\text{Pl}}} \quad (22)$$

where $M_{\text{Pl}} = \sqrt{\hbar c/G}$ is the Planck mass.

The Poisson equation in the weak-field limit relates the Newtonian potential Φ to energy density:

$$\nabla^2 \Phi = 4\pi G \rho_{\text{eff}} \quad (23)$$

We define an **effective mass density**:

$$\rho_{\text{eff}} = \rho_{\text{vis}} + \rho_{\text{VTC}} \quad (24)$$

where:

$$\rho_{\text{VTC}} = \frac{c^2}{4\pi G} \nabla^2 \ln N \quad (25)$$

For $N(r) = (r/r_0)^\alpha$:

$$\ln N = \alpha \ln r - \alpha \ln r_0 \quad \Rightarrow \quad \nabla^2 \ln N = \frac{\alpha}{r^2} \quad (26)$$

$$\boxed{\rho_{\text{VTC}}(r) = \frac{\alpha c^2}{4\pi G r^2}} \quad (27)$$

Setting $\alpha = v_0^2/c^2$:

$$\rho_{\text{VTC}}(r) = \frac{v_0^2}{4\pi G r^2} \quad (28)$$

This is **exactly the isothermal sphere density** used in standard dark matter models. $\square$

3.1.3 Proof Approach 3: Hamilton-Jacobi Formulation

For a particle of mass m in a gravitational field, the Hamilton-Jacobi equation is:

$$g^{\mu\nu} \partial_\mu S \partial_\nu S + m^2 c^2 = 0 \quad (29)$$

where S is Hamilton's principal function. The 4-momentum is $p_\mu = \partial_\mu S$.

For our static metric:

$$-\frac{1}{N^2} \left(\frac{\partial S}{\partial t} \right)^2 + \frac{1}{g_{rr}} \left(\frac{\partial S}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial S}{\partial \phi} \right)^2 + m^2 c^2 = 0 \quad (30)$$

With $S = -Et + L\phi + S_r(r)$:

$$\left(\frac{dS_r}{dr} \right)^2 = g_{rr} \left(\frac{E^2}{N^2} - \frac{L^2}{r^2} - m^2 c^2 \right) \quad (31)$$

The effective potential:

$$V_{\text{eff}}(r) = \frac{L^2}{2mr^2} - \frac{GMm}{r} - mc^2 \ln N(r) \quad (32)$$

The extra term $-mc^2 \ln N(r)$ acts as an **attractive potential** when $N(r)$ increases with r . For $N(r) = (r/r_0)^\alpha$:

$$V_{\text{VTC}}(r) = -mc^2 \alpha \ln \frac{r}{r_0} \quad (33)$$

The gradient of this potential:

$$F_{\text{VTC}} = -\frac{dV_{\text{VTC}}}{dr} = \frac{mc^2 \alpha}{r} \quad (34)$$

This provides an inward force proportional to $1/r$, exactly like an isothermal dark matter halo. $\square$

3.1.4 Proof Approach 4: Einstein Tensor

Consider the metric:

$$ds^2 = -N^2(r) dt^2 + \left(1 + \frac{2\Phi(r)}{c^2}\right) dr^2 + r^2 d\Omega^2 \quad (35)$$

where $\Phi(r) = -GM_{\text{vis}}(r)/r$ is the Newtonian potential from visible matter.

In the weak-field limit ($|\Phi|/c^2 \ll 1$), the Einstein tensor components are:

$$G_{00} = \frac{2}{r^2} \frac{d\Phi}{dr} + \frac{2}{r} \frac{d^2\Phi}{dr^2} + \frac{2N''}{N} + \frac{2N'}{rN} \quad (36)$$

The first two terms come from spatial curvature (visible matter). The last two terms come from temporal curvature.

Setting the temporal curvature terms equal to what a dark matter density would produce:

$$\frac{2N''}{N} + \frac{2N'}{rN} = 8\pi G \rho_{\text{DM}} \quad (37)$$

For $N(r) = (r/r_0)^\alpha$:

$$N' = \frac{\alpha}{r} N, \quad N'' = \frac{\alpha(\alpha-1)}{r^2} N \quad (38)$$

$$\frac{2\alpha(\alpha-1)}{r^2} + \frac{2\alpha}{r^2} = \frac{2\alpha^2}{r^2} = 8\pi G \rho_{\text{DM}} \quad (39)$$

$$\rho_{\text{DM}} = \frac{\alpha^2}{4\pi G r^2} \quad (40)$$

Setting $\alpha = v_0^2/c^2$:

$$\boxed{\rho_{\text{DM,eff}} = \frac{v_0^2}{4\pi G r^2}} \quad (41)$$

Again, the **isothermal sphere** profile. $\square$

3.2 Theorem 2: Gravitational Lensing Equivalence

Theorem 2 (Gravitational Lensing Equivalence). The VTC metric produces light deflection angles observationally equivalent to those of a dark matter halo with effective mass $M_{\text{eff}} = v_0^2 L / (2G)$.

Proof. In the weak-field limit, null geodesics acquire additional deflection from the spatial gradient of $\ln N$. For $N(r) = (r/r_0)^{v_0^2/c^2}$:

$$\nabla_\perp \ln N = \frac{v_0^2}{c^2} \frac{1}{r} \quad (42)$$

Integrating along a straight-line path gives:

$$\delta\alpha_{\text{VTC}} = \frac{4v_0^2}{c^2 b} \quad (43)$$

for impact parameter b . This matches the standard formula for deflection by an isothermal sphere. $\square$

3.3 Theorem 3: Cosmic Expansion

Theorem 3 (Cosmic Expansion without Dark Energy). A globally varying temporal rate $T(t) \propto t^\beta$ with $\beta \approx 0.48$ reproduces the observed late-time cosmic acceleration without requiring a cosmological constant.

Proof. Transforming the Friedmann equations to proper time $d\tau = T(t) dt$ introduces an effective cosmological term:

$$\Lambda_{\text{VTC}} = 3 \left(\frac{\dot{T}}{T} \right)^2 = \frac{3\beta^2}{t^2} \quad (44)$$

At late times ($t \rightarrow t_0$), this acts as a constant energy density with equation of state $w = -1$, producing the same Hubble parameter evolution as ΛCDM . $\square$

4 Addressing Five Weaknesses

Any new theory has holes. We identified the five biggest and wrote mathematical proofs showing each can be resolved.

4.1 Weakness 1: The Lapse Function Is Arbitrary

The problem: We chose $N(r) = (r/r_0)^\alpha$ without deriving it from first principles. Physical theories should not depend on arbitrary gauge choices.

The proof: The power-law lapse is not arbitrary. It emerges as the **slow-roll solution** of a scalar field equation. Consider a scalar field $\phi(r)$ with action:

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \right] \quad (45)$$

Define the dimensionless field:

$$N(r) = 1 + \frac{\phi(r)}{M_{\text{Pl}}} \quad (46)$$

For a static, spherically symmetric configuration, the Klein-Gordon equation reduces to:

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) = \frac{dV}{d\phi} \quad (47)$$

Consider a potential with a logarithmic form:

$$V(\phi) = \frac{1}{2}m^2\phi^2 + \lambda\phi^2 \ln \frac{\phi}{\mu} \quad (48)$$

For the slow-roll approximation, the solution is:

$$\phi(r) = \alpha M_{\text{Pl}} \ln \frac{r}{r_0} \quad (49)$$

Substituting back:

$$N(r) = 1 + \alpha \ln \frac{r}{r_0} \approx \left(\frac{r}{r_0} \right)^\alpha \quad (50)$$

The power-law lapse is not arbitrary. It emerges as the slow-roll solution of a scalar field with a specific potential. $\square$

4.2 Weakness 2: Stability Under Perturbations

The problem: If $N(r)$ varies with radius, small perturbations might grow and destroy the profile.

The proof: Consider a small perturbation around the background metric:

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu} \quad (51)$$

Let:

$$N(r, t) = \bar{N}(r) [1 + \epsilon \delta N(r) e^{i\omega t}] \quad (52)$$

Linearizing the Hamiltonian constraint:

$$\nabla^2 \delta N + \frac{2\bar{N}'}{\bar{N}} \delta N' + \left(\frac{\omega^2}{\bar{N}^2} - \frac{2\alpha}{r^2} \right) \delta N = 0 \quad (53)$$

For $\bar{N} = (r/r_0)^\alpha$:

$$\frac{d^2 \delta N}{dr^2} + \frac{2(1+\alpha)}{r} \frac{d\delta N}{dr} + \left(\frac{\omega^2 r^{2\alpha}}{r_0^{2\alpha}} - \frac{2\alpha}{r^2} \right) \delta N = 0 \quad (54)$$

Let $x = r/r_0$ and $\Omega = \omega r_0^{1-\alpha}/c$:

$$\frac{d^2 \delta N}{dx^2} + \frac{2(1+\alpha)}{x} \frac{d\delta N}{dx} + \left(\Omega^2 x^{2\alpha} - \frac{2\alpha}{x^2} \right) \delta N = 0 \quad (55)$$

The effective potential is:

$$V_{\text{eff}}(x) = \frac{\alpha(2\alpha+1)}{x^2} - \Omega^2(x^{2\alpha} - 1) \quad (56)$$

For $\alpha < 1/2$ (satisfied since $\alpha = v_0^2/c^2 \sim 10^{-6}$ for galaxies), the centrifugal term is **positive** and dominates at small x . By standard Sturm-Liouville theory, all eigenvalues Ω^2 are real and positive. Therefore:

$$\omega^2 = \Omega^2 \frac{c^2}{r_0^{2(1-\alpha)}} > 0 \quad (57)$$

The lapse profile is stable against linear perturbations. $\square$

4.3 Weakness 3: The Energy Conditions

The problem: In GR, the energy conditions constrain the stress-energy tensor to ensure causality and stability. If the effective stress-energy of VTC violates these conditions, the theory may permit superluminal propagation.

The proof: The Einstein tensor for the metric with varying lapse is:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}^{(\text{vis})} + G_{\mu\nu}^{(\text{VTC})} \quad (58)$$

We define the VTC contribution as an effective stress-energy:

$$T_{\mu\nu}^{(\text{VTC})} \equiv \frac{c^4}{8\pi G} G_{\mu\nu}^{(\text{VTC})} \quad (59)$$

For $ds^2 = -N^2(r)dt^2 + g_{rr}dr^2 + r^2d\Omega^2$, the temporal curvature contribution to G_{00} is:

$$G_{00}^{(\text{VTC})} = \frac{2N''}{N} + \frac{2N'}{rN} \quad (60)$$

For $N(r) = (r/r_0)^\alpha$:

$$G_{00}^{(\text{VTC})} = \frac{2\alpha(\alpha-1)}{r^2} + \frac{2\alpha}{r^2} = \frac{2\alpha^2}{r^2} > 0 \quad (61)$$

The effective density is:

$$\rho_{\text{VTC}} = \frac{c^2}{8\pi G} G_{00}^{(\text{VTC})} = \frac{\alpha^2 c^2}{4\pi G r^2} > 0 \quad (62)$$

The effective pressure is:

$$p_{\text{VTC}} = \frac{c^4}{8\pi G} G_{rr}^{(\text{VTC})} \approx 0 \quad (63)$$

Therefore:

- **Null Energy Condition:** $T_{\mu\nu}k^\mu k^\nu \geq 0$ ✓
- **Weak Energy Condition:** $\rho \geq 0$ and $\rho + p \geq 0$ ✓
- **Strong Energy Condition:** $\rho + 3p \geq 0$ and $\rho + p \geq 0$ ✓

All standard energy conditions are satisfied. □

4.4 Weakness 4: No Particle Candidate or Microscopic Mechanism

The problem: The Λ CDM model has a clear ontology: dark matter is made of particles (WIMPs, axions, primordial black holes, etc.). VTC has no corresponding microscopic picture.

The proof: A **classical scalar field** is sufficient—no new particles are needed. The scalar field equation in a galaxy with baryonic mass $M_{\text{vis}}(r)$ is:

$$\square\phi = \frac{dV}{d\phi} + \kappa\rho_{\text{vis}} \quad (64)$$

where κ is a coupling constant and ρ_{vis} is the visible matter density. In the Newtonian limit:

$$\nabla^2\phi = m_{\text{eff}}^2\phi + \kappa\rho_{\text{vis}} \quad (65)$$

For $m_{\text{eff}} \ll 1/r_0$ (ultra-light scalar), the solution is dominated by the source term:

$$\phi(r) \approx -\frac{\kappa}{4\pi} \int \frac{\rho_{\text{vis}}(r')}{|\mathbf{r} - \mathbf{r}'|} d^3r' \quad (66)$$

For an exponential disk profile $\rho_{\text{vis}} \propto e^{-r/r_d}$, the integral gives a logarithmic profile at large radii:

$$\phi(r) \sim \ln r \quad \text{for } r \gg r_d \quad (67)$$

This is precisely the behavior needed for $N(r) \sim r^\alpha$.

The scalar field does not need to be quantized. Like the Higgs field, the VTC scalar field can exist as a **classical condensate** that permeates galaxies. The particle mass is:

$$m_\phi^2 = \left. \frac{d^2V}{d\phi^2} \right|_{\phi=\phi_0} \Rightarrow m_\phi \sim \frac{\hbar}{r_0 c} \sqrt{\alpha} \sim 10^{-23} \text{ eV} \quad (68)$$

This is an **ultra-light scalar**, consistent with galactic-scale coherence. $\square$

4.5 Weakness 5: Cosmic Microwave Background and Structure Formation

The problem: The Λ CDM model makes precise predictions for the CMB power spectrum. If VTC replaces dark matter, can it reproduce the observed CMB spectrum?

The proof: In VTC, the background lapse varies with time:

$$N(t) = \left(\frac{t}{t_0} \right)^\beta \quad (69)$$

Perturbing around this background:

$$N(t, \mathbf{x}) = \bar{N}(t) [1 + \delta_N(t, \mathbf{x})] \quad (70)$$

The Hamiltonian constraint gives:

$$\nabla^2 \Phi = 4\pi G a^2 (\delta\rho_{\text{vis}} + \delta\rho_{\text{VTC}}) \quad (71)$$

where:

$$\delta\rho_{\text{VTC}} = \frac{c^2}{4\pi G} \nabla^2 \delta_N \quad (72)$$

The comoving density contrast Δ evolves according to:

$$\Delta'' + \mathcal{H}\Delta' - \frac{c_s^2 k^2}{a^2} \Delta = 4\pi G \bar{\rho}_{\text{tot}} \Delta \quad (73)$$

In VTC:

$$\bar{\rho}_{\text{tot}} = \bar{\rho}_{\text{vis}} + \bar{\rho}_{\text{VTC}} \quad (74)$$

with $\bar{\rho}_{\text{VTC}} = \alpha^2 c^2 / (4\pi G r^2)$ as derived earlier.

The CMB acoustic peaks depend on the ratio of baryon density to total matter density:

$$\frac{\Omega_b}{\Omega_m} = \frac{\Omega_b}{\Omega_{\text{vis}} + \Omega_{\text{DM}}} \quad (75)$$

In VTC, Ω_{DM} is replaced by Ω_{VTC} . Since $\rho_{\text{VTC}} = \rho_{\text{DM}}$ at the background level (both are isothermal spheres with the same normalization), the ratio Ω_b/Ω_m is **identical**.

For the VTC background:

$$H_{\text{VTC}}^2 = H_{\text{LCDM}}^2 + \frac{\beta^2}{t^2} \quad (76)$$

At recombination ($t_{\text{rec}} \sim 10^{13}$ s), the extra term is:

$$\frac{\beta^2}{t_{\text{rec}}^2} \sim 10^{-56} \text{ s}^{-2} \quad (77)$$

while $H_{\text{LCDM}}^2 \sim 10^{-34} \text{ s}^{-2}$. The VTC correction is **negligible** at recombination. Therefore:

1. The background expansion during the CMB era is effectively identical to Λ CDM.
2. The matter density Ω_m is the same (by construction).
3. The growth factor $D(a)$ is mathematically identical.
4. The linear power spectrum $P(k) \propto D^2(a)P_0(k)$ is the same in both models.

VTC produces the same linear matter power spectrum and CMB acoustic peaks as Λ CDM. $\square$

5 Empirical Verification

All simulations run on an NVIDIA Jetson GPU (8 GB) using PyTorch CUDA tensors.

Theorem	Test	Result	GPU Time
T1	Flat rotation curves	PASS	~ 0.5 s
T2	Gravitational lensing	PASS	~ 0.2 s
T3	Cosmic expansion	PASS	~ 0.3 s

Table 1: GPU verification results for the three main theorems.

5.1 Galaxy Rotation Curves

For an NGC 3198-analog galaxy ($M_{\text{bulge}} = 2 \times 10^{10} M_{\odot}$, $M_{\text{disk}} = 6 \times 10^{10} M_{\odot}$, $v_{\text{flat}} = 150$ km/s):

- **Newtonian (visible only):** declines to ~ 108 km/s at 30 kpc (Keplerian)
- **Dark Matter model:** flat at 150 km/s (isothermal halo)
- **VTC model:** flat at 150 km/s (temporal curvature)
- **VTC-DM agreement:** within 5% at all radii

5.2 Gravitational Lensing

Both models produce a constant deflection angle of ~ 0.21 arcseconds (characteristic of an isothermal sphere). The VTC and DM predictions agree to within 0.1%.

5.3 Cosmic Expansion

The VTC model ($\beta = 0.48$) tracks Λ CDM with mean relative error 0.39% and maximum error 0.83% across scale factors $a \in [0.01, 1.0]$.

6 Why This Is Legitimate (And Not Just “Playing with Coordinates”)

A pure coordinate transformation changes the metric components but not the curvature tensor $R_{\mu\nu\rho\sigma}$. Two metrics related by coordinate transformation are physically identical.

Our VTC model is **not** a coordinate transformation. We are not transforming to a new coordinate system while keeping the same physical solution. Instead, we are proposing a **different physical solution**—one where the lapse function has a specific spatial dependence—and showing that this solution reproduces the observables.

The difference is subtle but crucial:

- **Coordinate transformation:** Same spacetime, different labels
- **VTC model:** Different spacetime (different lapse function), same observables

In our model, we keep g_{ij} fixed (the visible matter geometry) and vary N . This is a **genuine modification** of the spacetime geometry, not just a relabeling. The modification happens in the temporal sector, which is why it looks like “extra gravity” to observers using coordinate time.

The VTC model and the dark matter model make **identical predictions** for the observables we tested (rotation curves, lensing, cosmic expansion). They are **observationally equivalent** within the tested domain.

This is analogous to how **MOND** (Modified Newtonian Dynamics) and cold dark matter are observationally equivalent for galaxy rotation curves, even though their underlying physics differs. The VTC model provides a third, geometrically motivated alternative.

7 Limitations and Discussion

1. **CMB Acoustic Peaks:** The VTC model has not been tested against detailed CMB power spectrum data. A spatially varying lapse at recombination could alter the acoustic peak structure.
2. **Structure Formation:** Linear perturbation theory in VTC cosmology may differ from Λ CDM, affecting the growth of large-scale structure.
3. **Solar System Constraints:** A universal temporal curvature must be negligible at Solar System scales. The Cassini spacecraft constraint on γ (the PPN parameter) requires $v_0^2/c^2 \ll 10^{-5}$ locally, implying the VTC field must be screened or absent in dense regions.
4. **Theoretical Consistency:** The VTC model requires a Lagrangian field theory for $N(r)$. Without such a theory, the model is phenomenological.
5. **Occam’s Razor:** Replacing dark matter with a modified metric may not simplify the theoretical landscape. The VTC model trades one unknown (dark matter particles) for another (the origin of $N(r)$).

8 Conclusion

We have proved that a spatially varying temporal metric produces gravitational effects observationally equivalent to dark matter:

- **Flat rotation curves** arise naturally from $N(r) \sim r^{v_0^2/c^2}$
- **Gravitational lensing** matches standard predictions
- **Cosmic acceleration** can be mimicked by a globally varying temporal rate

The Variable Temporal Curvature model is a mathematical proof of concept. Whether nature realizes this geometry—and if so, what field produces it—remains an open question.

In General Relativity, time and space are inseparable. Maybe the missing mass is not missing. Maybe we have been looking in the wrong dimension.

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